```
# Using enumerate()
# enumerate() function allows you to loop through a container while getting both the index and the


words = ['The', 'Big', 'Bang', 'Theory']
for index, value in enumerate(words):
    print(index, value)
```

```
0 The
1 Big
2 Bang
3 Theory
```

```
# Using zip()
# zip() function combines two or more containers and iterates over them in parallel. The loop stop

names = ['Leo', 'Kendall', 'Harry']
ages = (24, 27, 25)

for name, age in zip(names, ages):
    print(f"Name: {name}, Age: {age}")
```

```
Name: Leo, Age: 24
Name: Kendall, Age: 27
Name: Harry, Age: 25
```

```
# Using items()
# items() allows you to loop through a dictionary and access both its keys and values simultaneous

p1 = {'Washington': 'First', 'Lincoln': 'Emancipator', 'Roosevelt': 'Trust-Buster'}
for name, title in p1.items():
    print(name, title)
```

```
Washington First
Lincoln Emancipator
Roosevelt Trust-Buster
```

```
# Using sorted()
# sorted() iterates over the list in ascending order without changing the original list.

num = [1, 3, 5, 6, 2, 1, 3]

print("Sorted list:")
for n in sorted(num):
    print(n, end=" ")

print("\nSorted list without duplicates:")
for n in sorted(set(num)):
    print(n, end=" ")
```

```
Sorted list:
1 1 2 3 3 5 6
Sorted list without duplicates:
1 2 3 5 6
```

```
# Using reversed()
# reversed() iterates over a container in reverse order without modifying it.

num = [1, 3, 5, 6, 2, 1, 3]

for n in reversed(num):
    print(n, end=" ")
```

```
3 1 2 6 5 3 1
```

```
# While Loop with If Condition
# while loops are useful when you need conditional iteration.
count = 0

while count < 5:
    if count == 3:
        print("Count is 3")
    count += 1
```

```
Count is 3
```